

Project Design Phase Solution Architecture

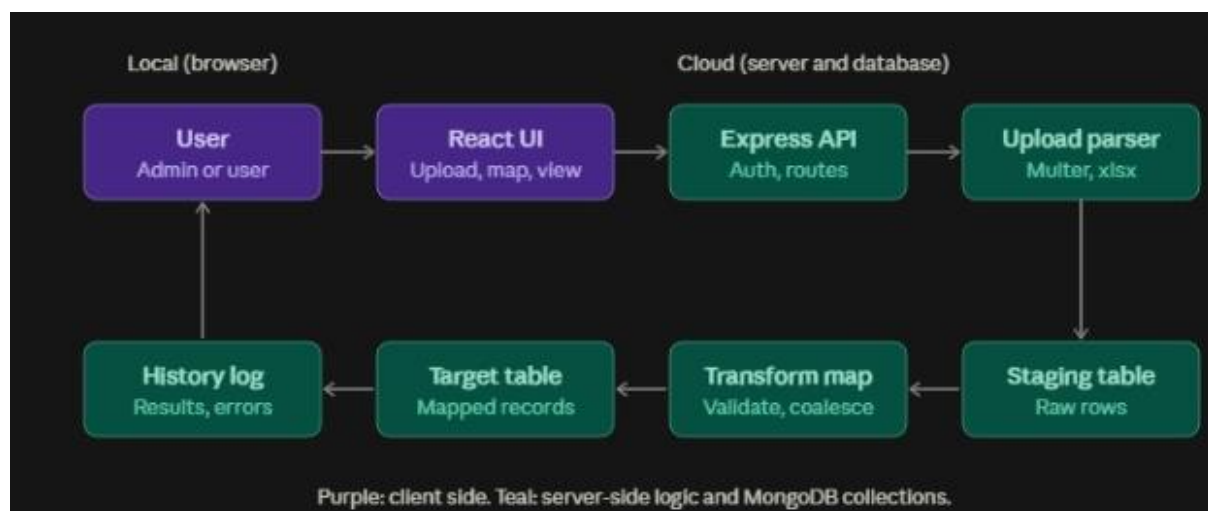
Date	01 October 2026
Team ID	SWTID-2026-2085
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Solution Architecture

The solution bridges a common business problem, slow and error-prone manual data entry from spreadsheets, and a technology solution: a transform-map-based import system. Its goals are to:

- Find the best technical solution for bulk spreadsheet data entry, using a staging table and a configurable transform map.
- Describe the structure and behavior of the software to project stakeholders: a React client, a Node.js/Express server, and MongoDB storage.
- Define features, development phases, and solution requirements: upload, mapping, validation, coalesce, import, and history.
- Provide specifications by which the solution is defined, managed, and delivered: REST APIs, database collections, and role-based access.

Example - Solution Architecture Diagram:



Architecture and Data Flow

Figure 1: Architecture and data flow of the spreadsheet import application (diagram above).

1. The user logs in and uploads an .xlsx/.csv file through the React UI.
2. The Express API checks the JWT token and the user's role.
3. The upload parser reads the file and saves the raw rows into the staging table.
4. The transform map matches source columns to target fields, validates the data, and applies the coalesce rule to avoid duplicates.
5. Valid records are inserted or updated in the target table.
6. The history log stores the inserted, updated, and failed counts, and the results are returned to the user in the UI.